KMTC/QP-08/EPS

KENYA MEDICAL TRAINING COLLEGE
FACULTY OF INFORMATION TECHNOLOGY AND MEDICAL EDUCATION
DEPARTMENT OF HEALTH RECORDS AND INFORMATION

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

PAPER: HEALTH STATISTICS

DATE: 12th JULY, 2021	**TIME: 3 HOURS (9:00AM – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. Which of the following is not a measure of central tendency

   a) Mean
   b) Median
   c) Mode
   d) Skewness

2. When plotting the pie chart the angles should equal to

   a) 360^0
   b) 100^0
   c) 170^0
   d) A and B

3. Given Q1 as 400 and Q3 as 520 which of the following is the inter-quartile deviation

   a) 40
   b) 60
   c) 50
   d) 65

4. Which of the following information presentation methods is best suitable for economic financial distribution?

   a) Lorenz curve
   b) Time series
   c) Pie chart
   d) Percentage component bar chart

5. Aloete medical centre with 10 available beds rendered 480 patient days in June 2019. What was the percentage occupancy for hospital for the month of June 2019?

   a) 79.9%
   b) 150%
   c) 160%
   d) 90%

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6. The first quartiles divide a frequency distribution in the ratio of

a) 4:1
b) 1:4
c) 3:1
d) 1:3

7. The following are measures of dispersion except

a) Mean
b) Mean deviation
c) Skewness
d) Standard deviation

8. A table that shows multiple columns and is used to measure a change of one variable in two different times is known as

a) Contingency table
b) Compound table
c) Simple table
d) Component table

9. The self-discharge of a patient without following the formal discharge process is

a) To transfer
b) To parole
c) To abscond
d) To referral

10. Facts and figures collected together for reference or analysis are called

a) Data
b) Statistics
c) Information
d) Presentation

11. A distribution where mean is less than the median and medium is less than the mode is termed to be

a) Skewed to the right
b) Symmetrical
c) Skewed to the left
d) Asymmetrical

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12. Which two major variables do we measure in a research process?

a) Qualitative and quantitative
b) Independent and qualitative
c) Dependent and quantitative
d) Independent and dependent

13. Data collected from journals is known as

a) Discrete data
b) Sampling
c) Primary
d) Secondary

14. Which of the following is not categorical data

a) Hair colour
b) Opinion of the people
c) Education level
d) Smooth status

15. The number of times that a score occurs is referred to as

a) Array distribution
b) Frequency distribution
c) Score distributions
d) Frequency

16. Which of the following does not refer to allocated beds

a) Assigned beds
b) Bed management
c) Staffed beds
d) Available beds

17. ..... is a value that is calculated for a group of data and it should be a representation of all the values and becomes the average

a) Measures of statistics
b) Measures of central tendency
c) Measures of variation
d) Measures of dispersion

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18. Which of the following does not refer to median

a) The eleven value
b) 50th percentile
c) Middle value
d) Value dividing distribution into two equal parts

19. which measure of central tendency is affected by outliers

a) Mean
b) Median
c) Mode
d) Skewness

20. Which type of variables is characterized by gaps or interruption?

a) Discrete random
b) Continuous random
c) Qualitative
d) Quantitative

21. P x Q would always be

a) Less than one
b) More than one
c) Zero
d) One

22. Additional theorem states that if two events A and B are mutually exclusive, the probability of occurrence of either A and B is given by

a) P(A) + P(B)
b) P(B) x P(A)
c) P(A) – P(B)
d) P(A) x P(B)

23. If two evens A and B are dependent, the conditional probability of A given B is calculated as

a) P(B/A) (AB)
b) P(B/P(A))
c) P(AB/P(A))
d) P(AB)/P(B)

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24. The standard duration of binomial distribution is

a) √nap
b) √npq
c) Npq-npq
d) Np-np

25. When variables are not measurable, which of the following is to be represented

a) Nominal
b) Ordinal
c) Interval
d) Ratio

26. Which of the following is not a discrete variate?

a) Skin colour
b) Blood pressure
c) Weight
d) Boys in class

27. Which of the following types of diagrams can be used to find out the relationship between two variables?

a) Pictogram
b) Bar diagram
c) Histogram
d) Scatter diagram

28. A pictorial diagram of frequency distribution is denoted by

a) Line chart
b) Pictogram
c) Histogram
d) Pie chart

29. The "real limits" of the class interval 2-4 are

a) 2.5 and 4.5
b) 2 and 4
c) 1.5 and 4.5
d) 1.5 and 3.5

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30. If mean, median and mode are 10, 16 and 22 respectively, the distribution is

a) Symmetric
b) Normal
c) Positively skewed
d) Negatively skewed

31. If "V: stands for variation and "S" stands for standard deviation, then the relation between V and S is

a) V=S
b) $V=\sqrt{S}$
c) $V^2 = S$
d) $S^2 = V$

32. The ratio of 3:1 is achieved by

a) Quartile
b) Median
c) 3rd quartile
d) 1st quartile

33. Area between one standard deviation on either side of mean in normal distribution curve is

a) 50%
b) 68%
c) 136%
d) 95%

34. A type I error is committed when?

a) False H_0 is accepted
b) True H_0 is rejected
c) True H_1 is accepted
d) False H_1 is rejected

35. P-valve is the probability of

a) Rejecting H_0 when it is true
b) Accepting H_0 when it is true
c) Rejecting H_0 when it is false
d) Rejecting H_1 when it is true

36. Which of the following is unrelated to the chi-square test of significance

a) Degree of freedom
b) Life table
c) Significance level
d) Qualitative data

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37. Which of the following statements is true about simple random sampling

a) Techniques that provide least number of possible samples
b) Every fixed unit is taken for selection
c) All units have equal chance to be selected
d) Only selected units have right to be selected

38. for a survey, a village is divided into 5 lanes then each is sampled randomly. It is an example of

a) Simple random sampling
b) Stratified random sampling
c) Systematic random sampling
d) Multi-stage random sampling

39. Median is almost equivalent to

a) 25th percentile
b) 50th percentile
c) 75th percentile
d) 10th percentile

40. The semi interquartile range is most closely related to

a) Mean
b) Median
c) Mode
d) Range

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. State five principles of questionnaire construction (5 marks)
42. Describe the stratified sampling method approach (5 marks)
43. Outline five characteristics of statistical data (5 marks)
44. Describe five application of vital statistics in health sector (5 marks)
45. Describe five uses of Lorenze curve (5 marks)
46. In the year 2018, Kocholia sub-county hospital had 800 beds permanently allocated for in-patient use during the same period the percentage occupancy was 110% and 1500 patients were discharged home alive while 20 died. Calculate

(i) Turn over per bed (3 marks)
(ii) Available bed days ABD (2 marks)

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47. Outline 5 important points to consider while constructing a histogram (5 marks)

48. Find the area under normal curve less than 1.64 (5 marks)

SECTION THREE: LONG ANSWER QUESTIONS (20 MARKS)

49. A random of 50 male employees is taken at the end of a year and the mean number of hou of absenteeism for the year is found to be 63 hours. A similar sample of 50 female employees has a mean of 66 hours. Could these samples be drawn from a population with the same mean and standard deviation of 10 hours? (10 marks)

50. In 2001, Moding Health Centre recorded 11,676 inpatients days, 130 discharges and death and the health facility has a capacity of 30 beds. Calculate (10 marks)

    (i) ABD (1 mark)
    (ii) EVBD (1 mark)
    (iii) Percentage occupancy (1 mark)
    (iv) Average length of stay (ALOS) (2 marks)
    (v) Supposing this unit represent 25% of the entire health facility bed and inpatients days. Calculate the entire health facility statistics. (5 marks)

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